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EVA-CLIP-18B: Scaling CLIP to 18 Billion Parameters

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code & models: baaivision/EVA/EVA-CLIP-18B

Abstract

Scaling up contrastive language-image pretraining (CLIP) is critical for empowering both vision and multimodal models. We present ***EVA-CLIP-18B****, the largest and most powerful open-source CLIP model to date, with 18-billion parameters. With only 6-billion training samples seen,* ***EVA-CLIP-18B*** *achieves an exceptional* ***80.7%*** *zero-shot top-1 accuracy averaged across 27 widely recognized image classification benchmarks, outperforming its forerunner* ***EVA-CLIP*** *(5-billion parameters) and other open-source CLIP models by a large margin. Remarkably, we observe a consistent performance improvement with the model size scaling of* ***EVA-CLIP****, despite maintaining a constant training dataset of 2-billion image-text pairs from LAION-2B and COYO-700M. This dataset is openly available and much smaller than the in-house datasets (e.g., DFN-5B, WebLI-10B) employed in other state-of-the-art CLIP models.* ***EVA-CLIP-18B*** *demonstrates the potential of EVA-style [30, 29, 63] weak-to-strong visual model scaling. With our model weights made publicly available, we hope to facilitate future research in vision and multimodal foundation models.*

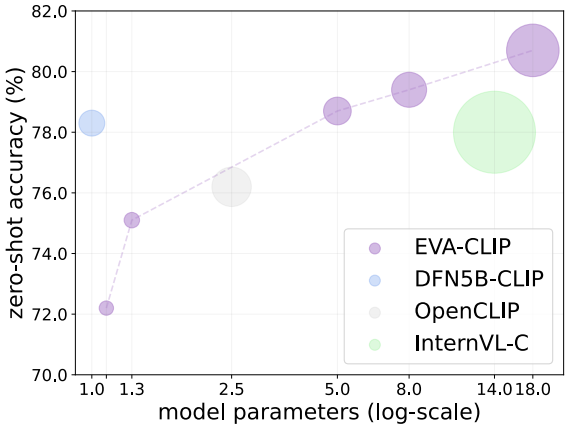

Figure 1: **Scaling behavior of EVA-CLIP with zero-shot classification performance averaged across 27 image classification benchmarks**, compared with the current state-of-the-art and largest CLIP models (224px). The diameter of each circle demonstrates the forward GFLOPs × the number of training samples seen. The performance of EVA-CLIP consistently improves as scaling up.

1. Introduction

Recent years witnessed the rapid growth of Large Multimodal Models (LMMs) [3, 64, 62, 69, 5, 46], with CLIP models [53, 19, 63, 43, 75, 28, 17] serving as a foundational vision encoder to deliver robust and transferable visual representations, and Large Language Models (LLMs) [65, 54] serving as a general interface for reasoning across different modalities. However, as LLMs have scaled up to around 100B parameters or higher [11, 20, 65], the adopted vision foundation models continue to operate at a much smaller scale, lagging far behind the LLMs.

This paper introduces EVA-CLIP-18B, the largest open-source CLIP model with 18-billion parameters to narrow this gap. EVA-CLIP [63] open-sources a series of effective and efficient CLIP models, which have been leveraged as the vision foundation by numerous impactful works across 2D / 3D vision and multimodal modeling [42, 78, 77, 50, 69, 64]. We further scale up EVA-CLIP to this significant parameter size building upon the scaling philosophy of EVA [30, 29] and EVA-CLIP [63]. With merely 6-billion training samples seen and trained on publicly available datasets, EVA-CLIP-18B achieves the exceptional **80.7%** average zero-shot top-1 accuracy on 27 widely recognized image classification benchmarks, significantly surpassing its forerunner EVA-02-CLIP-E/14+ (5-billion parameters) and other open-source CLIP models. Besides, the models *have not exhibited any*

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model	total #param.	image #param.	text #param.	data	samples seen	image size	batch size	gpus for training	image cls. avg. acc.	video cls. avg. acc.	retrieval MR
EVA-01-CLIP-g/14 [63]	1.1B	1.0B	124M	LAION-400M [59]	11B	224^2	41k	256×A100 (40GB)	72.2	66.2	80.9
EVA-01-CLIP-g/14+ [63]	1.3B	1.0B	354M	Merged-2B [63]	11B	224^2	114k	112×A100 (40GB)	75.1	68.8	85.3
OpenCLIP-G/14 [2]	2.5B	1.8B	695M	LAION-2B [58]	39B	224^2	160k	512×A100 (80GB)	76.2	68.7	85.7
InternVL-C [17]	14.0B	6.0B	8.0B	custom [17]	29B	224^2	164k	640×A100 (80GB)	78.0	73.7	86.6
DFN5B-CLIP-H/14 [28]	1.0B	632M	354M	DFN-5B [28]	39B	224^2	79k	TPUv4	78.3	67.0	86.6
EVA-02-CLIP-E/14+ [63]	5.0B	4.4B	695M	LAION-2B [58]	9B	224^2	144k	144×A100 (80GB)	78.7	72.1	85.7
DFN5B-CLIP-H/14+ [28]	1.0B	632M	354M	DFN-5B [28]	5B	378^2	79k	TPUv4	79.2	68.4	87.2
EVA-CLIP-8B	8.1B	7.5B	695M	Merged-2B [63]	9B	224^2	178k	384×A100 (40GB)	79.4	73.6	86.2
EVA-CLIP-18B	18.1B	17.5B	695M	Merged-2B+	6B	224^2	108k	360×A100 (40GB)	**80.7**	**75.0**	**87.8**

Table 1: **CLIP model configurations and zero-shot performance on 33 benchmarks including 27 image classification, 4 video classification and 2 image-text retrieval datasets.** DFN-5B [28] are 5B images filtered from a pool of 43B uncurated image-text pairs consisting of 12.8B image-text pairs from CommonPool-12.8B [32] and 30B additional public image-text pairs. The dataset used for training InternVL-C [17] is custom mixtures, see detail in [17].

method	ImageNet-1K [26]	ImageNet-V2 [57]	ImageNet-Adv. [36]	ImageNet-Ren. [35]	ImageNet-Ske. [68]	ObjectNet [8]	CIFAR-10 [40]	CIFAR-100 [40]	MNIST [41]	Caltech101 [31]	SUN397 [72]	FGVC Aircraft [47]	Country-211 [53]	Stanford Cars [39]	Birdsnap [9]	DTD [21]	EuroSAT [34]	FER2013 [33]	Flowers-102 [49]	Food-101 [10]	GTSRB [61]	PCam [67]	Pets [51]	Rendered SST2 [53]	RESISC45 [18]	STL-10 [23]	VOC2007 [27]	**avg. top-1 acc.**
EVA-01-CLIP-g/14	78.5	71.5	73.6	92.5	67.6	72.3	98.3	88.7	62.6	87.7	74.2	32.4	28.9	91.7	65.8	61.7	73.8	52.2	74.5	93.5	49.3	49.9	94.2	58.4	70.3	98.9	85.7	72.2
EVA-01-CLIP-g/14+	79.3	72.5	74.1	92.7	68.4	75.3	99.1	90.1	72.0	89.5	74.7	39.9	31.8	90.7	70.2	67.8	73.2	56.0	79.7	93.7	66.5	62.4	94.9	58.6	71.4	99.5	84.7	75.1
OpenCLIP-G/14	80.4	73.6	69.3	92.8	69.9	73.0	98.3	87.5	71.6	89.4	75.0	53.6	34.9	94.9	73.0	69.1	71.1	**59.6**	81.5	93.1	62.7	63.6	95.3	65.3	72.6	98.5	87.4	76.2
InternVL-C	83.2	77.3	83.8	**95.7**	74.3	80.6	**99.4**	93.1	80.6	89.5	76.3	53.3	35.1	94.4	69.2	70.8	79.4	56.2	85.8	95.3	65.5	48.7	96.3	**68.4**	74.4	99.4	80.0	78.0
DFN5B-CLIP-H/14	83.5	77.4	71.7	92.9	72.8	76.7	98.8	90.5	**85.8**	89.5	77.0	71.4	34.4	95.8	77.4	70.7	65.2	54.7	**92.5**	95.8	67.7	65.2	96.5	54.8	76.1	98.9	81.5	78.3
EVA-02-CLIP-E/14+	82.1	75.7	82.1	94.7	72.2	79.6	99.3	93.2	74.7	**90.5**	75.3	58.7	37.0	94.7	77.6	68.2	75.9	59.0	84.5	94.9	67.7	64.4	96.0	62.6	75.7	99.3	**87.9**	78.7
DFN5B-CLIP-H/14+	**84.3**	**78.3**	79.6	93.6	73.3	79.6	98.8	90.5	83.6	88.9	77.4	**72.5**	37.9	**96.0**	**80.5**	70.9	61.1	56.1	91.6	**96.2**	67.9	**69.6**	**96.8**	55.5	75.9	99.1	81.9	79.2
EVA-CLIP-8B	83.5	77.7	85.2	95.3	74.3	81.2	99.3	92.3	84.8	89.6	76.2	60.5	41.7	94.8	79.0	71.0	68.9	56.1	86.4	95.5	70.9	58.1	96.4	66.2	75.3	99.3	85.1	79.4
EVA-CLIP-18B	83.8	77.9	**87.3**	**95.7**	**74.7**	**82.2**	**99.4**	**93.8**	83.0	89.8	**77.7**	59.7	**43.1**	94.9	79.9	**72.1**	**79.8**	59.3	86.0	95.8	**72.4**	65.2	96.1	67.5	**76.9**	**99.6**	85.8	**80.7**

Table 2: EVA-CLIP **zero-shot image classification performance on 27 datasets.** We report top-1 accuracy on all datasets. The best results are in **bold** and the second best are underlined.

signal of performance saturation, shedding light on further scaling of vision models. An intuitive demonstration is shown in Figure 1.

The successful training of EVA-CLIP-18B exemplifies the potential of the EVA-style visual model scaling philosophy. We keep open-sourcing the training code and weights of our models to encourage further research and empower the development of vision and multimodal foundation models.

2. Weak-to-Strong Vision Scaling

Our scaling-up procedure is guided by the principles of EVA [30] and EVA-CLIP [63]. The EVA philosophy for scaling visual models follows a weak-to-strong paradigm, designed to improve visual models through a strategic progression. This process begins with a large EVA vision model distilling knowledge from a small EVA-CLIP model, which in turn serves as the vision encoder initialization to stabilize and accelerate the training of a larger EVA-CLIP. After that, the closed-loop scaling-up cycle continues and a larger EVA is distilled out. Throughout our model scaling cycle, the training dataset remains largely fixed to demonstrate the effectiveness of our model-scale specific scaling philosophy, although scaling up datasets can further unleash the scaling power of our method.

Specifically, in this work, we pre-train a large EVA model named EVA-18B using a small EVA-CLIP (EVA-02-CLIP-E/14+) [63] as the teacher, which is trained to reconstruct masked image-text aligned vision features from EVA-02-CLIP-E/14+. EVA-18B omits bias terms of QKV projections and uses RMSNorm [76] instead of LayerNorm [4] following LLaMA [65]. Subsequently, we leverage the EVA model as the vision encoder initialization for EVA-CLIP pre-training with the image-text contrastive learning objective. Besides, we also introduce a smaller counterpart, EVA-CLIP-8B, which undergoes similar pre-training methodologies. Notably, our experiments demonstrate sustained performance improvement with the progressive weak-teach-strong scaling up of EVA-CLIP.

3. Experiments

Settings. Following EVA-CLIP [63], we initialized the model with pre-trained vision and text encoders. Specifi-

method	zero-shot **text** retrieval						zero-shot **image** retrieval						MR
	Flickr30K			COCO			Flickr30K			COCO			
	R@1	R@5	R@10	R@1	R@5	R@10	R@1	R@5	R@10	R@1	R@5	R@10	
EVA-01-CLIP-g/14	87.9	98.0	99.5	61.7	83.2	89.9	72.5	91.5	95.4	44.5	69.1	77.7	80.9
EVA-01-CLIP-g/14+	93.3	99.5	99.9	69.4	88.3	93.2	79.2	95.2	97.3	51.1	74.7	82.5	85.3
OpenCLIP-G/14	93.5	99.3	99.7	69.0	87.8	93.1	80.9	95.1	97.2	52.6	76.1	83.6	85.7
EVA-02-CLIP-E/14+	94.3	99.6	99.8	69.4	88.6	93.3	79.7	94.9	97.3	52.5	75.9	83.4	85.7
EVA-CLIP-8B	95.6	99.6	99.9	70.3	89.3	93.9	80.8	95.5	97.6	53.0	76.0	83.4	86.2
DFN5B-CLIP-H/14	92.9	99.3	99.9	72.3	90.2	94.6	80.1	95.2	97.3	53.9	78.0	85.6	86.6
InternVL-C	93.8	**99.7**	**100.0**	70.3	89.2	93.8	82.1	96.0	98.1	54.1	77.1	84.8	86.6
DFN5B-CLIP-H/14 +	93.6	99.3	99.6	71.8	90.4	94.9	82.1	96.0	97.9	55.6	**79.2**	**86.3**	87.2
EVA-CLIP-18B	**96.7**	**99.7**	**100.0**	**73.6**	**90.9**	**95.0**	**83.3**	**96.3**	**98.3**	**56.2**	78.5	85.6	**87.8**

Table 3: Zero-shot retrieval performance on Flickr30K [74] and COCO [45].

cally, we employ a pre-trained EVA-18B [30, 29] as the vision encoder and EVA-02-CLIP-E/14+ [63] for the text encoder. We adopt the LAMB optimizer [73] with $\beta_1 = 0.9$, β_2=0.95, and a weight decay of 0. We apply different learning rates and layer decay rates to the vision encoder and text encoder to ensure optimal training. We set the peak learning rate as 4e-4 and 4e-5 for the vision encoder and the text encoder respectively, with 2000 warm-up steps. Afterwards, the learning rates decay to 0 with a cosine schedule. The learning rate layer decay rates are configured as 0.9 and 0.75 for the vision and text encoders. The temperature parameter remains constant at 0.01. Further, we use the `DeepSpeed` optimization library [56] with ZeRO stage-3 partition [55], gradient checkpointing [16] and flash attention [24] to optimize the training cost.

Dataset. Our Merged-2B dataset consists of 1.6 billion samples from LAION-2B [58] and 0.4 billion samples from COYO-700M [12]. Note that the use of a subset from LAION-2B is not the result of deliberate filtering, but rather due to image downloading failures. The use of 0.4 billion COYO-700M samples aims to complement the number of training samples to nearly the same as LAION-2B. Merged-2B+ consists of all samples from Merged-2B, along with additional 20 million samples from LAION-COCO [1] and 23 million samples from Merged-video including VideoCC [48], InternVid [70] and WebVid-10M [6]. Merged-video is included at the end of the training process.

EVA-CLIP-18B pre-trains with 5.4 billion samples from Merged-2B seen with 50% of patch dropout ratio [44], 0.6 billion samples from Merged-2B and 20 million samples from LAION-COCO without patch dropout, and 24 million samples from Merged-video with 50% of patch dropout ratio.

Evaluation. We evaluate on 33 widely used datasets across image, video classification and image-text retrieval. All datasets used to evaluate EVA-CLIP-18B are reported in Table 11. We utilize the specified prompt templates following [53, 38].

method	image encoder			text encoder			# params		
	layers	width	heads	layers	width	heads	image	text	total
EVA-CLIP-8B	32	4096	32	32	1280	20	7.5B	695M	8.1B
EVA-CLIP-18B	48	5120	40	32	1280	20	17.5B	695M	18.1B

Table 4: Architecture configurations.

Zero-Shot Image Classification. We show the exceptional performance of EVA-CLIP on all 27 zero-shot image classification benchmarks in Table 2. EVA-CLIP-18B achieves **80.7%** top-1 accuracy averaged across all 27 benchmarks. These results significantly outperform the previous best open-sourced DFN5B-CLIP-H/14+ [28] by **+1.5%**, and the largest existing CLIP model, InternVL-C [17], by **+2.7%**. For Birdsnap dataset, the download was limited to 2195 test images due to broken links.

method	#Frames	UCF-101	K-400	K-600	K-700	**avg.**
EVA-01-CLIP-g/14	1	76.0	65.4	64.5	58.8	66.2
DFN5B-CLIP-H/14	1	78.2	65.2	65.5	59.2	67.0
DFN5B-CLIP-H/14+	1	79.2	66.7	67.0	60.7	68.4
OpenCLIP-G/14	1	80.5	67.1	66.9	60.3	68.7
EVA-01-CLIP-g/14+	1	78.9	67.3	67.3	61.5	68.8
EVA-02-CLIP-E/14+	1	83.1	70.7	70.0	64.4	72.1
EVA-CLIP-8B	1	85.7	71.3	71.2	66.1	73.6
InternVL-C	1	85.2	71.8	71.7	66.4	73.7
EVA-CLIP-18B	1	**86.0**	**72.9**	**72.9**	**68.2**	**75.0**
EVA-CLIP-18B	8	**88.2**	**79.3**	**79.2**	**72.1**	**79.7**
EVA-CLIP-18B	16	**88.4**	**79.4**	**79.4**	**72.2**	**79.8**

Table 5: EVA-CLIP **zero-shot video classification performance.** We report top1 accuracy for UCF-101 [60], average of top1 and top5 accuracy for Kinetics-400 [15], Kinetics-600 [13] and Kinetics-700 [14].

Zero-Shot Video Classification. We report the top-1 accuracy for UCF-101 [60] and the mean of top-1 and top-5 accuracy for Kinetics-400 [15], Kinetics-600 [13] and Kinetics-700 [14]. In Table 5 we demonstrate that EVA-CLIP-18B also outperforms other CLIP models on zero-shot video clas-

method	IN-1K	IN-A	IN-R	IN-V2	IN-Sketch	ObjectNet	Δ↓	avg. acc.
DFN5B-CLIP-H/14	83.5	71.7	92.9	77.4	72.8	76.7	4.4	79.2
OpenCLIP-G/14	80.4	69.3	92.8	73.6	69.9	73.0	3.9	76.5
SigLIP-SO [75] (reported)	82.0	71.9	95.1	76.1	74.0	70.6	3.7	78.3
DFN5B-CLIP-H/14+	84.3	79.6	93.6	78.3	73.3	79.6	2.8	81.5
EVA-01-CLIP-g/14	78.5	73.6	92.5	71.5	67.6	72.3	2.5	76.0
EVA-01-CLIP-g/14+	79.3	74.1	92.7	72.5	68.4	75.3	2.2	77.1
BASIC-L [52] (reported)	85.7	85.6	95.7	80.6	76.1	82.3	1.4	84.3
SigLIP-SO+ [75] (reported)	83.0	82.5	95.8	77.2	74.5	77.0	1.3	81.7
EVA-02-CLIP-E/14+	82.1	82.1	94.7	75.7	72.2	79.6	1.0	81.1
InternVL-C	83.2	83.8	95.7	77.3	74.3	80.6	0.7	82.5
EVA-CLIP-8B	83.5	85.2	95.3	77.7	74.3	81.2	0.6	82.9
EVA-CLIP-18B	83.8	87.3	95.7	77.9	74.7	82.2	**0.2**	83.6

(a) **Zero-shot performance on ImageNet variants and ObjectNet.** "avg. acc.": the averaged top-1 accuracy on different ImageNet variants (*i.e.*, IN-{1K, V2, ReaL, Adv., Ren., Ske.}), and ObjectNet. "Δ↓": The gap between the averaged top-1 accuracy and the ImageNet-1K top-1 accuracy (the lower the better). EVA-CLIP suffers from the smallest performance drop (only 0.2% top-1 accuracy gap for EVA-CLIP-18B) while EVA-CLIP-18B achieves 83.6% top-1 accuracy averaged on all 6 benchmarks.

method	ImageNet-1K [26]	ImageNet-V2 [57]	ImageNet-Adv. [36]	ImageNet-Ren. [35]	ImageNet-Ske. [68]	ObjectNet [8]	CIFAR-10 [40]	CIFAR-100 [40]	MNIST [41]	SUN397 [72]	Birdsnap [9]	DTD [21]	EuroSAT [34]	Food-101 [10]	PCam [67]	RESISC45 [18]	STL-10 [23]	**avg. top-1 acc.**
BASIC-L [52] (reported)	85.7	80.6	85.6	95.7	76.1	82.3	97.5	82.3	40.3	76.2	59.2	64.6	51.0	95.1	59.6	72.7	99.6	76.7 (77.8)
EVA-CLIP-18B	83.8	77.9	87.3	95.7	74.7	82.2	99.4	93.8	83.0	77.7	79.9	72.1	79.8	95.8	65.2	76.9	99.6	84.9 (84.1)
	-1.9	**-2.7**	**+1.7**	+0.0	**-1.4**	**-0.1**	**+1.9**	**+11.5**	**+42.7**	**+1.5**	**+20.7**	**+7.5**	**+28.8**	**+0.7**	**+5.6**	**+4.2**	+0.0	**+8.2** (**+6.3**)

(b) **Comparison EVA-CLIP-18B's zero-shot image classification performance with BASIC-L [52] on 17 datasets.** Our report includes the top-1 accuracy for all datasets, considering that BASIC-L only provided top-1 accuracy for these specific 17 datasets. ( ) is the average top-1 accuracy removing Birdsnap due to the different test size between EVA-CLIP-18B and BASIC-L. EVA-CLIP-18B outperforms BASIC-L with a notable margin of +8.2 (+6.3) in average top-1 accuracy, despite exhibiting lower performance on ImageNet variants.

Table 6: **Robustness evaluation of CLIP models and comparison with BASIC-L [52] on 17 Benchmarks.**

sification benchmarks by a large margin. When sampling a single center frame per video, EVA-CLIP-18B achieves accuracies of 86.0%, 72.9%, 72.9%, and 68.2% across the four evaluated benchmarks. Further, when uniformly sample 8 or 16 frames per video, we observe an improvement of +4.7% / +4.8% averaged across four benchmarks compared to the single-frame setting.

Zero-Shot Image-Text Retrieval. In Table 3, we report the zero-shot image and text retrieval results on Flickr30K [74] and COCO [45]. EVA-CLIP-18B achieves an average recall of 87.8% across all retrieval benchmarks, significantly outperforming competitors.

Robustness. In Table 6, we demonstrate that scaling up EVA-CLIP significantly enhances the robustness of visual representations. EVA-CLIP suffers from the smallest performance drop (Δ↓) between ImageNet-1K and ImageNet variants including adversarial ones, with merely 0.2% top-1 accuracy gap for EVA-CLIP-18B.

For a more robust and comprehensive evaluation of robustness and zero-shot performance, it is advisable to include more benchmarks covering more image distributions. However, we want to note that higher ImageNet top-1 accuracy does not necessarily lead to better overall performance, as evidenced in Table 6b, where BASIC-L [52] exhibits higher ImageNet-related top-1 accuracy but considerably lower overall average top-1 accuracy compared to EVA-CLIP-18B across a broader range of datasets and distributions, showing a difference of -8.2%.

Linear Probing on ImageNet-1K. In Table 7, we present the results of linear probing on ImageNet-1K [26]. EVA-CLIP-18B achieves an average top-1 accuracy of 88.9%, surpassing InternVL-C [17] by 0.7%.

method	#param	top1 acc.
OpenCLIP-G/14 (reported)	1.8B	86.2
EVA-01-CLIP-g/14	1.0B	86.5
EVA-02-CLIP-E/14+	4.4B	88.1
InternVL-C (reported)	5.9B	88.2
EVA-CLIP-8B	7.5B	88.5
EVA-CLIP-18B	17.5B	88.9

Table 7: **Linear Probing on ImageNet-1K [26].** The top-1 accuracy shows a continuous improvement with the scaling up of EVA-CLIP.

3D Representation. We adopt the Uni3D [77] setting to explore the effectiveness of scaling up teachers. With the scaling up of EVA-CLIP in Table 8, we observe consistent improvements in 3D representation learning capabilities. Further, Uni3D-base equipped with EVA-CLIP-18B sets new records on ModelNet [71] and ScanObjectNN [66] benchmarks.

teacher	data	O-LVIS	MNet40	ScanObjNN
OpenCLIP-G/14	w/o LVIS	44.5	85.8	58.9
EVA-02-CLIP-E/14+	w/o LVIS	45.8	86.1	61.7
EVA-CLIP-8B	w/o LVIS	46.2	87.3	62.7
EVA-CLIP-18B	w/o LVIS	**47.0**	**87.6**	**65.3**
EVA-02-CLIP-E/14+	Ensembled	51.7	86.3	63.8
EVA-CLIP-18B	Ensembled	**53.2**(+1.5)	**88.6**(+2.3)	**67.8**(+4.0)

Table 8: **EVA-CLIP-18B enhances zero-shot 3d classification performance.** We use Uni3D-base [77] as the baseline and scale the teacher from 5B to 18B. We report top-1 accuracy on Objaverse-LVIS [25], ModelNet40 [71] and ScanObjectNN [66].

4. Ablation Studies

Video Data. In Table 9, we conduct ablations on EVA-CLIP-18B's zero-shot performance, comparing results when trained with and without Merged-Video. The training objective for the video data aligns with that of images, encompassing the extraction of features from video where 8 frames are uniformly sampled. The mean of all [CLS] embeddings serves as a representation for the video. The outcomes reveal substantial performance improvements associated with training using Merged-Video. The zero-shot performance, averaged across UCF-101 [60] and Kinetics-400 [15] / 600 [13] / 700 [14], indicates a gain of **+0.7** for evaluation with one middle frame and **+0.8** for evaluation with 8 frames.

	classification			retrieval
	image	video (#F 1)	video (#F 8)	avg. recall
w/o video data	80.7	74.3	78.9	87.9
w/ video data	80.7	75.0 (+0.7)	79.7 (+0.8)	87.8 (-0.1)

Table 9: **Video data enhances zero-shot video classification performance**. We respectively report performances averaged on 27 image classification benchmarks, 4 video benchmarks and 2 image-text retrieval benchmarks.

Image Resolution. In Table 10, we investigate the impact of larger image resolutions on zero-shot performance. Notably, there is an average top-1 accuracy gain of **+0.9** when the resolution increases from 224^2 to 448^2 for EVA-CLIP-8B. Similarly, an increase from 224^2 to 336^2 results in a gain of **+0.5**, even when trained with low global batch sizes of 24k for EVA-CLIP-8B + and 23k for EVA-CLIP-18B+.

method	resolution	IN-1K	IN-A	IN-R	IN-V2	IN-Ske.	ObjectNet	**avg.**
EVA-CLIP-8B	224×224	83.5	85.2	95.3	77.7	74.3	81.2	82.9
EVA-CLIP-8B +	448×448	83.8	88.7	95.4	77.7	74.1	82.9	83.8
		+0.3	**+3.5**	**+0.1**	+0.0	**-0.2**	**+1.7**	**+0.9**
EVA-CLIP-18B	224×224	83.8	87.3	95.7	77.9	74.7	82.2	83.6
EVA-CLIP-18B+	336×336	83.9	88.9	95.6	78.2	74.3	83.6	84.1
		+0.1	**+1.6**	**-0.1**	**+0.3**	**-0.4**	**+1.4**	**+0.5**

Table 10: **Increasing resolution.** We report zero-shot performance on ImageNet variants and ObjectNet.

5. Conclusion

We present EVA-CLIP-18B, the currently largest and most performant open-sourced CLIP model with 18-billion parameters. We show that following EVA's weak-to-strong vision scaling principle, we can further scale up CLIP models to a new record and advance SOTA on multiple prevalent benchmarks across image, video and 3D domains. Importantly, we demonstrate that scaling up the size of EVA-CLIP models consistently boosts performance with no sign of saturation, shedding light on future vision model scaling.

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EVA-CLIP-18B: Scaling CLIP to 18 Billion Parameters

Supplementary Material

Dataset	Classes	Test size	Evaluation Metric
ImageNet-1K [26]	1000	50,000	accuracy
ImageNet-V2 [57]	1000	10,000	accuracy
ImageNet-Adversarial [36]	1000	7,500	accuracy
ImageNet-R(endition) [35]	1000	30,000	accuracy
ImageNet-Sketch [68]	1000	50,899	accuracy
ObjectNet [8]	1000	50,273	accuracy
CIFAR-10 [40]	10	10,000	accuracy
CIFAR-100 [40]	100	10,000	accuracy
MNIST [41]	10	10,000	accuracy
Caltech101 [31]	101	9144	accuracy
SUN397 [72]	397	108,754	accuracy
FGVC Aircraft [47]	100	3,333	accuracy
Country-211 [53]	211	21,100	accuracy
Stanford Cars [39]	196	8,041	accuracy
Birdsnap [9]	500	2,195	accuracy
Describable Textures [21]	47	1,880	accuracy
EuroSAT[34]	10	27,000	accuracy
Facial Emotion Recognition 2013 [33]	8	3,574	accuracy
Oxford Flowers 102 [49]	102	6,149	accuracy
Food-101 [10]	102	25,250	accuracy
GTSRB [61]	43	12,630	accuracy
PatchCamelyon [67]	2	32,768	accuracy
Oxford-IIIT Pets [51]	37	3,669	accuracy
Rendered SST2 [53]	2	1,821	accuracy
RESISC45 [18]	45	31,500	accuracy
STL-10 [23]	10	8000	accuracy
Pascal VOC 2007 Classification [27]	20	4,952	accuracy
UC-F101 [60]	101	11,213	accuracy
Kinetics-400 [15]	400	19,240	mean(top1, top5)
Kinetics-600 [13]	600	29,788	mean(top1, top5)
Kinetics-700 [14]	700	33,966	mean(top1, top5)
Flickr30K [74]	-	1000	recall
COCO [45]	-	5000	recall

Table 11: Datasets used to evaluate EVA-CLIP models.

config	EVA-CLIP-{8B, 8B+}
image enc. weight init.	EVA-8B / EVA-CLIP-8B
text enc. weight init.	EVA-02-CLIP-E/14+ / EVA-CLIP-8B
image-text data	Merged-2B
image enc. peak learning rate	4e-4 / 2e-4
image enc. layer-wise lr decay [22, 7]	0.9 / 0.85
text enc. peak learning rate	4e-5 / 2e-5
text enc. layer-wise lr decay [22, 7]	0.75
learning rate schedule	cosine decay
optimizer	LAMB [73]
optimizer hyper-parameters	$\beta_1, \beta_2, \epsilon$ = 0.9, 0.95, 1e-6
weight decay	0
input resolution	224^2 / 448^2
patch size	14^2
batch size	178k / 24k
samples seen	9B / 800M
drop image patch [44]	0.5 / 0.0
drop path [37]	0.0
random resized crop	(0.9, 1)
numerical precision	`DeepSpeed bf16` [56]
ZeRO optimizer [55]	stage 3

Table 12: EVA-CLIP-8B and EVA-CLIP-8B+ training settings.

config	EVA-CLIP-{18B,18B+}
image enc. weight init.	EVA-18B / EVA-CLIP-18B
text enc. weight init.	EVA-02-CLIP-E/14+ / EVA-CLIP-18B
image-text data	Merged-2B+
image enc. peak learning rate	4e-4 / 2e-4
image enc. layer-wise lr decay [22, 7]	0.9 / 0.85
text enc. peak learning rate	4e-5 / 2e-5
text enc. layer-wise lr decay [22, 7]	0.75
learning rate schedule	cosine decay
optimizer	LAMB [73]
optimizer hyper-parameters	$\beta_1, \beta_2, \epsilon$ = 0.9, 0.95, 1e-6
weight decay	0
input resolution	224^2 / 336^2
patch size	14^2
batch size	108k / 23k
samples seen	6B / 400M
drop image patch [44]	0.5 / 0.0
drop path [37]	0.0
random resized crop	(0.9, 1)
numerical precision	`DeepSpeed bf16` [56]
ZeRO optimizer [55]	stage 3

Table 13: EVA-CLIP-18B and EVA-CLIP-18B+ training settings.

6. Training Settings

We present detailed training settings of EVA-CLIP-8B and EVA-CLIP-18B in Tabs. 12 and 13.

7. Image Transformations for Evaluation

Two prevalent image transformations utilized in zero-shot evaluation are: 1) direct resizing of images to a fixed size, such as 224×224, and 2) resizing images based on the shortest side, followed by center cropping to achieve a fixed size. In Table 14, our study systematically investigates the impact of these two image transformations in zero-shot evaluations. Notably, there exists a significant performance gap between the two transformations, observed particularly in zero-shot image classification on ObjectNet [8] and VOC2007 [27], and zero-shot retrieval on Flickr30K [74] and COCO [45]. EVA-CLIP-18B shows robustness with almost the same average accuracy across different image transformations in zero-shot image/video classification.

For zero-shot image classification and video classification,

method	ImageNet-1K [26]	ImageNet-V2 [57]	ImageNet-Adv. [36]	ImageNet-Ren. [35]	ImageNet-Ske. [68]	ObjectNet [8]	CIFAR-10 [40]	CIFAR-100 [40]	MNIST [41]	Caltech101 [31]	SUN397 [72]	FGVC Aircraft [47]	Country-211 [53]	Stanford Cars [39]	Birdsnap [9]	DTD [21]	EuroSAT [34]	FER2013 [33]	Flowers-102 [49]	Food-101 [10]	GTSRB [61]	PCam [67]	Pets [51]	Rendered SST2 [53]	RESISC45 [18]	STL-10 [23]	VOC2007 [27]	**avg. top-1 acc.**
DFN5B-CLIP-H/14+*	84.0	77.8	**79.6**	92.9	72.4	**79.6**	98.8	90.5	83.6	88.7	77.0	64.9	36.1	95.7	**80.5**	70.9	61.1	56.1	**91.6**	96.1	67.8	69.6	96.7	55.5	75.9	99.1	78.2	78.5
DFN5B-CLIP-H/14+†	**84.3**	**78.3**	79.3	**93.6**	**73.3**	73.5	98.8	90.5	83.6	**88.9**	**77.4**	**72.5**	**37.9**	**96.0**	80.3	70.9	61.1	56.1	91.4	**96.2**	**67.9**	69.6	**96.8**	55.5	75.9	99.1	**81.9**	**78.9**
EVA-CLIP-18B *	83.7	**77.9**	**87.3**	95.6	74.4	**82.2**	99.4	93.8	83.0	89.4	77.5	58.4	41.8	94.9	**79.9**	71.9	79.8	59.3	85.9	**95.8**	**72.4**	65.2	96.0	67.5	76.8	99.6	82.4	**80.4**
EVA-CLIP-18B †	**83.8**	77.7	86.2	**95.7**	**74.7**	76.2	99.4	93.8	83.0	**89.8**	**77.7**	**59.7**	**43.1**	94.9	78.4	**72.1**	79.8	59.3	**86.0**	95.7	72.3	65.2	**96.1**	67.5	**76.9**	99.6	**85.8**	**80.4**

(a) **Impact of image transformations on zero-shot image classification performance.** Different transformations can significantly influence zero-shot image classification performance, particularly for ObjectNet [8]. EVA-CLIP-18B shows robustness with the same average top-1 accuracy across different image transformations.

method	#Frames	UCF-101	K-400	K-600	K-700	**avg. acc.**
DFN5B-CLIP-H/14+*	1	78.5	65.2	66.0	59.2	67.2
DFN5B-CLIP-H/14+†	1	**79.2**	**66.7**	**67.0**	**60.7**	**68.4**
EVA-CLIP-18B *	1	**86.0**	72.2	72.6	67.4	74.6
EVA-CLIP-18B †	1	85.6	**72.9**	**72.9**	**68.2**	**74.9**
EVA-CLIP-18B *	8	**88.2**	**79.3**	**79.2**	72.0	**79.7**
EVA-CLIP-18B †	8	87.9	79.2	79.1	**72.1**	79.6

(b) **Impact of image transforms on zero-shot video classification performance.**

	zero-shot **text** retrieval						zero-shot **image** retrieval						
	Flickr30K			COCO			Flickr30K			COCO			
method	R@1	R@5	R@10	R@1	R@5	R@10	R@1	R@5	R@10	R@1	R@5	R@10	MR
DFN5B-CLIP-H/14 +*	92.3	99.1	**99.7**	70.6	89.6	94.4	80.7	95.5	97.7	54.1	78.0	85.4	86.4
DFN5B-CLIP-H/14 +†	**93.6**	**99.3**	99.6	**71.8**	**90.4**	**94.9**	**82.1**	**96.0**	**97.9**	**55.6**	**79.2**	**86.3**	**87.2**
EVA-CLIP-18B *	95.4	99.5	99.8	72.8	89.7	94.3	83.2	95.9	97.8	55.6	77.9	85.3	86.7
EVA-CLIP-18B †	**96.7**	**99.7**	**100.0**	**73.6**	**90.9**	**95.0**	**83.3**	**96.3**	**98.3**	**56.2**	**78.5**	**85.6**	**87.8**

(c) **Impact of image transforms on zero-shot retrieval performance.**

Table 14: **Impact of image transformations on zero-shot evaluation.** † denotes the direct resizing of images to a fixed size, while * indicates resizing images based on the shortest side and subsequently center cropping to achieve a fixed size.

we present results obtained by selecting the best-performing transformation between the two. In the case of zero-shot retrieval tasks, we specifically choose the transformation that involves direct resizing of images to a fixed size.